New Reactions

1. Radical Allylic Halogenations (Section 14-2)

$$\mathrm{RCH_2CH{=}CH_2} \xrightarrow{\text{NBS, } \mathrm{CCl_4},\ h\nu} \mathrm{R\overset{\displaystyle Br}{\overset{|}{C}}HCH{=}CH_2} \; + \; \mathrm{RCH{=}CHCH_2Br}$$

$DH°$ of allylic C–H bond ≈ 87 kcal mol^{-1}

2. SN2S_N2 Reactivity of Allylic Halides (Section 14-3)

$$\mathrm{CH_2{=}CHCH_2X} \; + \; \mathrm{Nu}\mathbf{:}^{-} \xrightarrow{\text{Acetone}} \mathrm{CH_2{=}CHCH_2Nu} \; + \; \mathrm{X^-}$$

Faster than ordinary primary halides

3. Allylic Grignard Reagents (Section 14-4)

$$\mathrm{CH_2{=}CHCH_2Br} \xrightarrow{\mathrm{Mg,\ (CH_3CH_2)_2O}} \mathrm{CH_2{=}CHCH_2MgBr}$$

Can be used in additions to carbonyl compounds

4. Allyllithium Reagents (Section 14-4)

$$\mathrm{RCH_2CH{=}CH_2} \xrightarrow{\mathrm{CH_3CH_2CH_2CH_2Li,\ TMEDA}} \mathrm{R\ddot{C}HCH{=}CH_2\ Li^+}$$

pK_a of allylic C–H bond ≈ 40

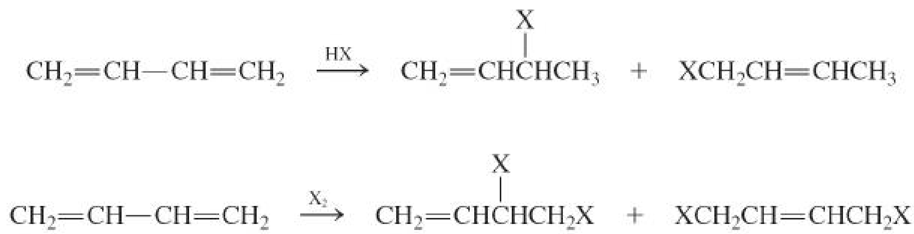
5. Hydrogenation of Conjugated Dienes (Section 14-5)

$$\mathrm{CH_2{=}CH{-}CH{=}CH_2} \xrightarrow{\mathrm{H_2,\ Pd\text{–}C,\ CH_3CH_2OH}} \mathrm{CH_3CH_2CH_2CH_3} \qquad \Delta H° = -57.1\ \text{kcal mol}^{-1}$$

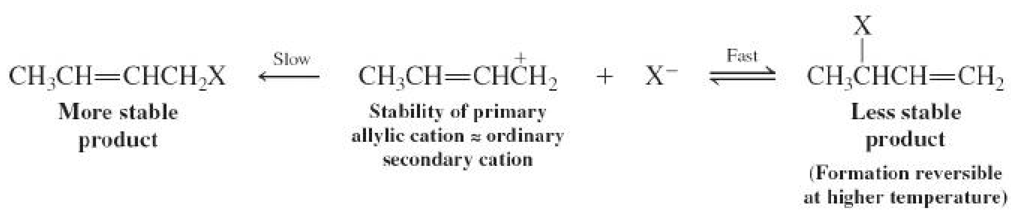
but compare

$$\mathrm{CH_2{=}CH{-}CH_2{-}CH{=}CH_2} \xrightarrow{\mathrm{H_2,\ Pd\text{–}C,\ CH_3CH_2OH}} \mathrm{CH_3(CH_2)_3CH_3} \qquad \Delta H° = -60.8\ \text{kcal mol}^{-1}$$

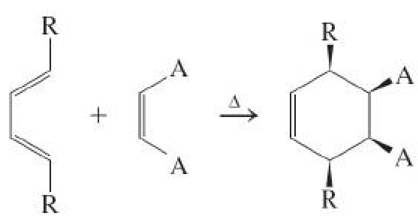
6. Electrophilic Reactions of 1,3-Dienes: 1,2- and 1,4-Addition (Section 14-6)



7. Thermodynamic Compared with Kinetic Control in SN1S_N1 Reactions of Allylic Derivatives (Section 14-6)

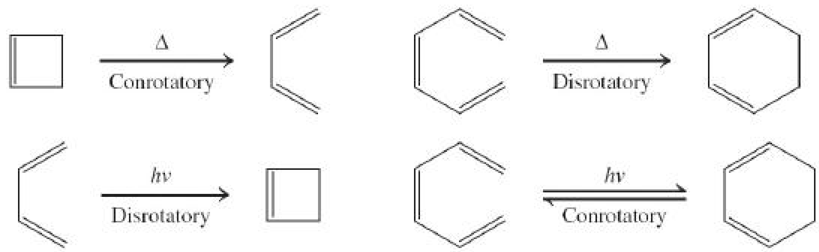


8. Diels-Alder Reaction (Concerted and Stereospecific, Endo Rule) (Section 14-8)



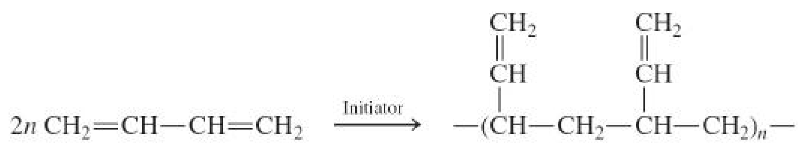
A = electron acceptor
Requires *s*-cis diene; better with electron-poor dienophile

9. Electrocyclic Reactions (Section 14-9)



10. Polymerization of 1,3-Dienes (Section 14-10)

1,2-Polymerization



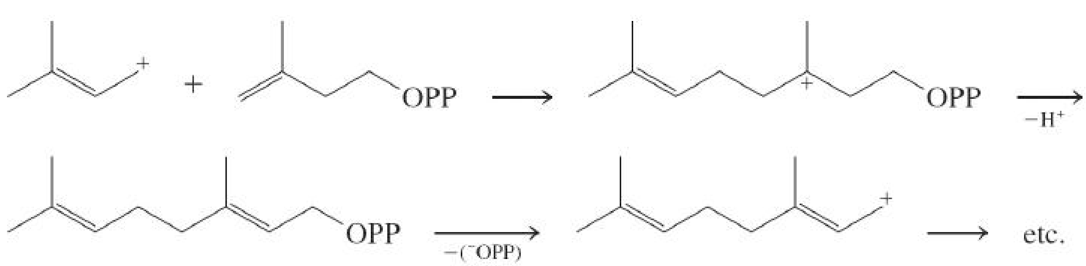
1,4-Polymerization

$$n\,CH_2{=}CH{-}CH{=}CH_2 \xrightarrow{\text{Initiator}} {-}(\underbrace{CH_2{-}CH{=}CH{-}CH_2}_{\text{Cis or trans}})_n{-}$$

11. 3-Methyl-3-butenyl Pyrophosphate as a Biochemical Building Block (Section 14-10)

$$\underset{\textbf{3-Methyl-3-butenyl pyrophosphate}}{CH_2{=}\overset{\overset{\large CH_3}{|}}{C}{-}CH_2CH_2OPP} \underset{}{\overset{\text{Enzyme}}{\rightleftharpoons}} (CH_3)_2C{=}CHCH_2OPP \longrightarrow \underset{\textbf{Allylic cation}}{(CH_3)_2C{=}CHCH_2{}^{+}} + \underset{\textbf{Pyrophosphate ion}}{{}^{-}OPP}$$

C–C bond formation



Important Concepts

1. The 2-propenyl **(allyl)** system is stabilized by **resonance.** Its molecular-orbital description shows the presence of three ππ molecular levels: one bonding, one nonbonding, and one antibonding. Its structure is symmetric, any charges or odd electrons being equally distributed between the two end carbons.
2. The chemistry of the 2-propenyl (allyl) cation is subject to both **thermodynamic and kinetic control.** Nucleophilic trapping may occur more rapidly at an internal carbon that bears relatively more positive charge, giving the thermodynamically less stable product. The kinetic product may rearrange to its thermodynamic isomer by dissociation followed by eventual thermodynamic trapping.
3. The stability of allylic radicals allows **radical halogenations** of alkenes at the allylic position.
4. The **SN2**S_N2 reaction of allylic halides is accelerated by orbital overlap in the transition state.
5. The special stability of allylic anions allows **allylic deprotonation** by a strong base, such as butyllithium-TMEDA.
6. 1,3-Dienes reveal the effects of **conjugation** by their relative stability (compared to nonconjugated systems) and a relatively short internal bond (1.47 Å).(1.47 Å).
7. Electrophilic attack on 1,3-dienes leads to the preferential formation of allylic cations.
8. **Extended conjugated systems** are reactive because they have many sites for attack and the resulting intermediates are stabilized by resonance.
9. Benzene has special stability because of **cyclic delocalization.**
10. The **Diels-Alder reaction** is a concerted stereospecific **cycloaddition reaction** of an *s*-cis diene to a dienophile; it leads to cyclohexene derivatives. It follows the **endo rule.**

11. Conjugated dienes and trienes equilibrate with their respective cyclic isomers by concerted and stereospecific **electrocyclic reactions.**

12. **Polymerization** of 1,3-dienes results in 1,2- or 1,4-additions to give polymers that are capable of further **cross-linking.** Synthetic rubbers can be synthesized in this way. Natural rubber is made by electrophilic carbon-carbon bond formation involving biosynthetic five-carbon cations derived from 3-methyl-3-butenyl pyrophosphate.

13. **Ultraviolet and visible spectroscopy** gives a way of estimating the extent of conjugation in a molecule. Peaks in **electronic spectra** are usually broad and are reported as λmax (nm).λ_{max} (nm). Their relative intensities are given by the **molar absorptivity** (extinction coefficient) $\in$.$\in$.